

SCPT Engineering Interpretation Report

Forward ray-path shear-wave velocity inversion, arrival-time comparison, and Vs30 summary

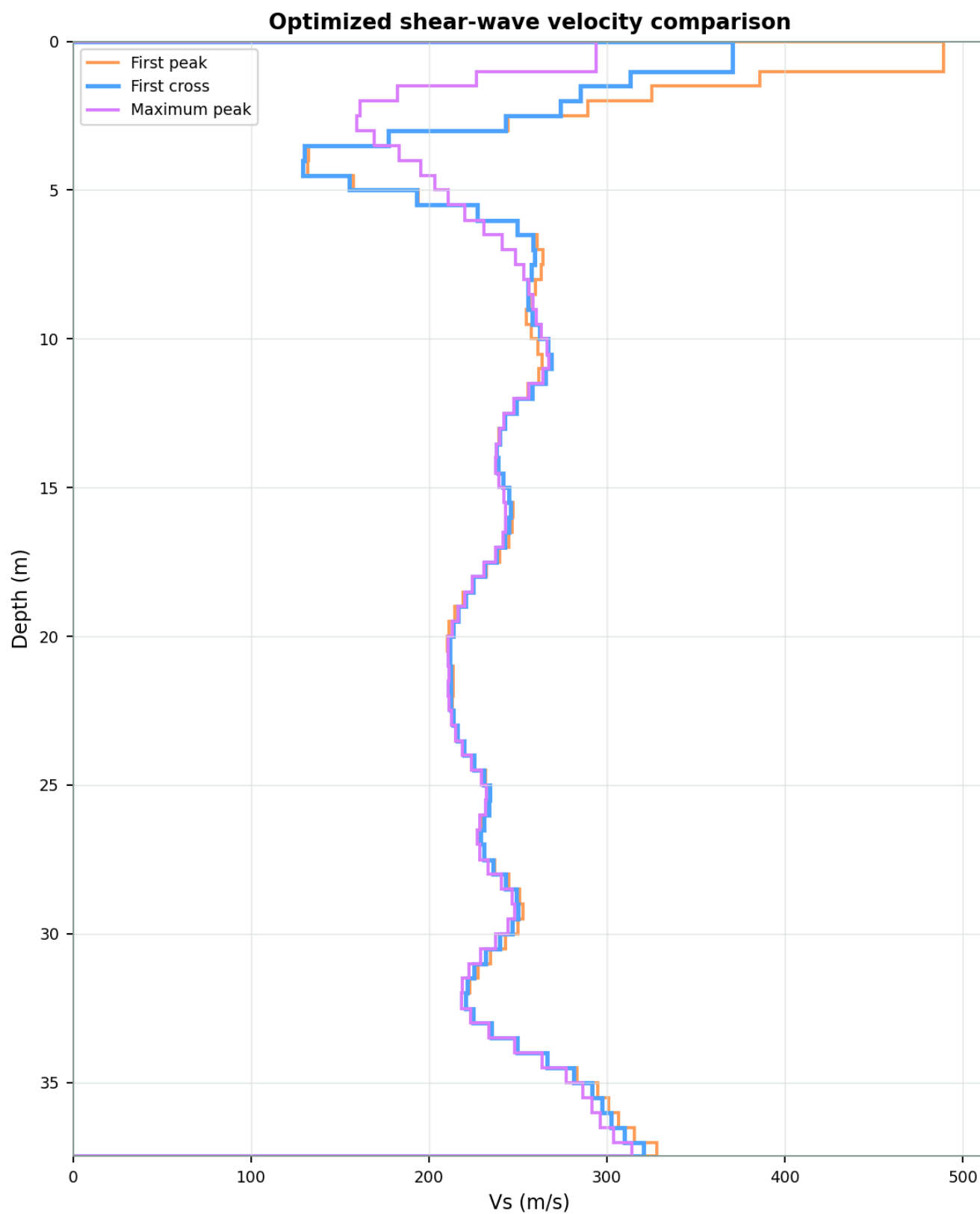
Report generated	2026-08-04 18:33 New Zealand Standard Time
Project	Test1.rpscpt
GRU source	18563_6.GRU
Source offset	2.400 m
Selected detailed model	First cross
Smoothing / regularisation	0.23
Vs30 extrapolation weighting	1.00
Deepest receiver	37.47 m
GRU pre-trigger correction	50.0 ms

Model comparison

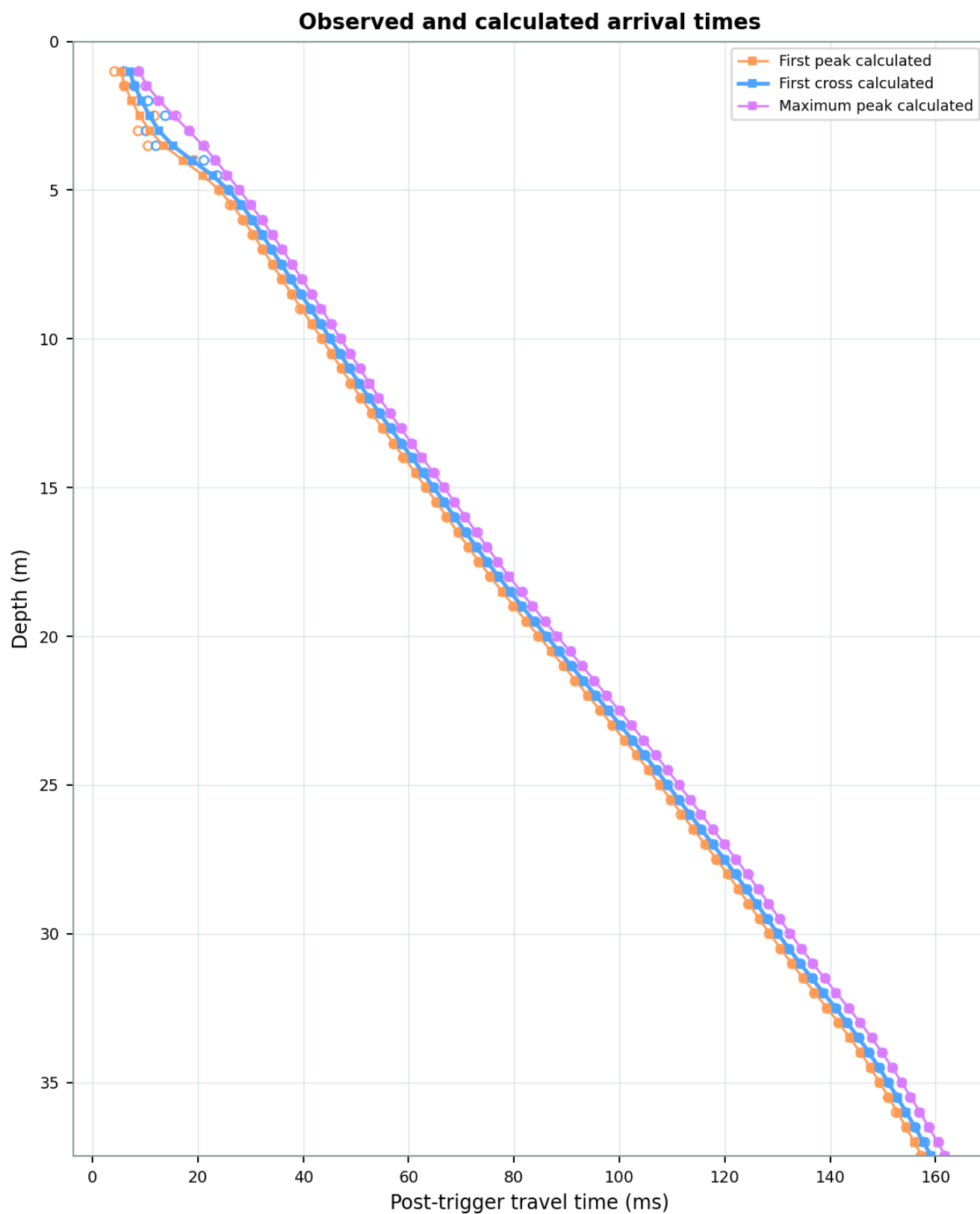
Pick definition	Layers	RMSE (ms)	Vs30 (m/s)	Extrapolation
First peak	74	0.648	234.1	None - measured to 30 m
First cross	74	0.685	231.3	None - measured to 30 m
Maximum peak	74	0.137	227.3	None - measured to 30 m

Vs30 is calculated as 30 divided by the summed vertical shear-wave travel time. Profiles between 25 m and 30 m use the configured interval weighting only to estimate the missing depth. Automatic waveform suggestions should be reviewed by a qualified operator before relying on this report.

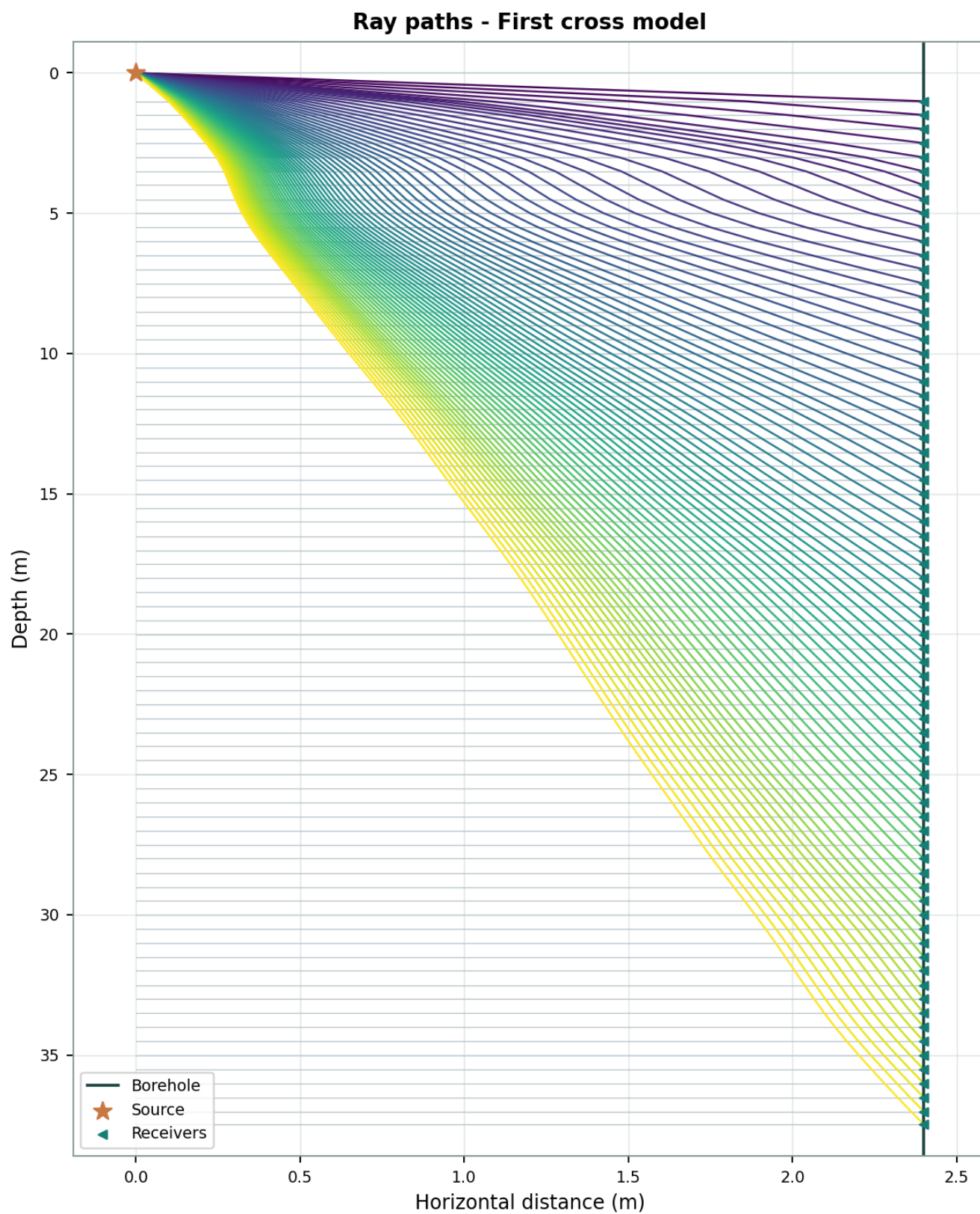
Velocity profile comparison



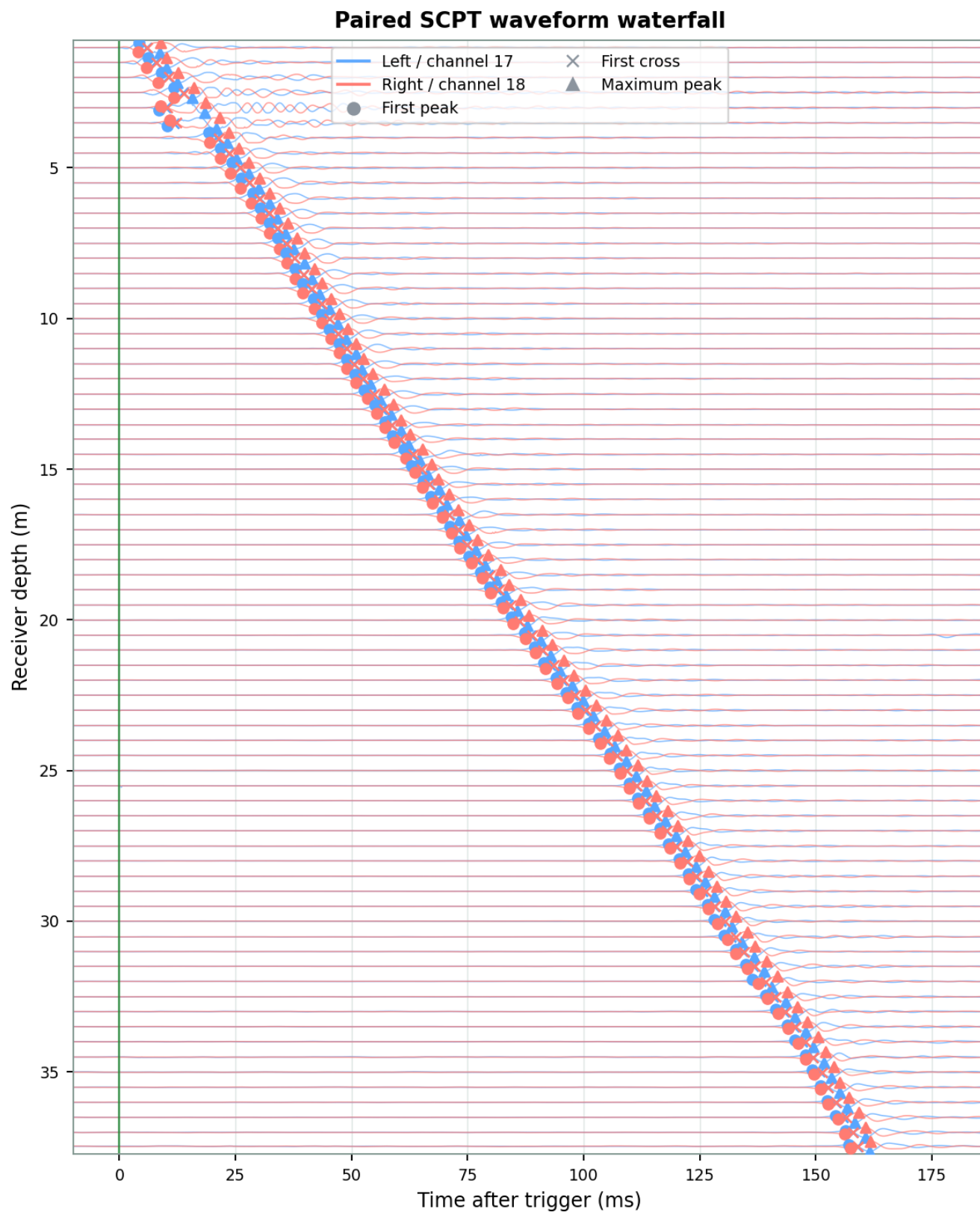
Arrival-time fit comparison



Selected-model ray paths



Waveform waterfall and reviewed picks



Receiver pick times

Depth (m)	First peak (ms)	First cross (ms)	Maximum peak (ms)
1.010	4.189	5.957	8.817
1.500	6.032	7.963	10.131
2.000	8.751	10.573	12.500
2.500	11.773	13.832	15.872
3.010	8.675	10.187	18.422
3.510	10.560	12.072	21.217
4.000	19.314	21.172	23.322
4.510	21.819	23.640	25.426
5.000	24.115	25.900	27.868
5.500	26.159	27.981	30.076
6.010	28.551	30.227	32.322
6.510	30.415	32.091	34.222
7.000	32.306	34.000	35.986
7.510	34.300	35.892	37.900
8.000	35.989	37.740	39.800
8.510	37.915	39.664	41.700
9.000	39.442	41.355	43.300
9.510	41.900	43.439	45.400
10.000	43.580	45.241	47.200
10.510	45.451	47.073	49.000
11.000	47.343	48.909	50.895
11.510	49.000	50.474	52.400
12.000	50.900	52.420	54.300
12.510	53.100	54.577	56.600
13.010	55.200	56.725	58.700
13.530	57.200	58.664	60.600
14.000	59.000	60.456	62.400
14.510	61.500	63.020	64.900
15.000	63.300	64.750	66.800
15.500	65.300	66.752	68.700
16.000	67.200	68.693	70.700
16.510	69.515	70.993	73.100
17.010	71.400	72.909	74.900
17.510	73.300	74.887	76.900
18.000	75.500	76.945	79.000
18.510	78.000	79.571	81.600
19.000	79.900	81.427	83.500
19.510	82.400	83.996	86.000
20.010	84.615	85.981	88.040

Depth (m)	First peak (ms)	First cross (ms)	Maximum peak (ms)
20.510	87.300	88.703	90.800
21.000	89.500	90.992	93.000
21.510	91.600	93.115	95.200
22.000	94.100	95.522	97.600
22.500	96.400	97.945	100.100
23.000	98.700	100.255	102.300
23.510	101.000	102.442	104.600
24.000	103.400	104.897	107.000
24.500	105.700	107.104	109.200
25.000	107.700	109.207	111.400
25.510	109.800	111.363	113.500
26.000	111.700	113.242	115.400
26.500	114.100	115.534	117.800
27.000	116.400	117.812	120.000
27.510	118.400	119.874	122.100
28.000	120.700	122.223	124.500
28.510	122.700	124.201	126.500
29.010	124.500	125.991	128.300
29.510	126.700	128.183	130.500
30.000	128.400	129.984	132.300
30.520	130.600	132.203	134.600
31.010	132.800	134.355	136.700
31.500	135.000	136.650	139.100
32.000	137.000	138.606	141.000
32.510	139.400	141.098	143.600
33.000	141.600	143.247	145.700
33.500	143.900	145.489	148.000
34.000	145.800	147.408	149.900
34.510	147.796	149.363	151.859
35.010	149.400	151.088	153.600
35.510	151.000	152.607	155.200
36.010	152.500	154.255	156.900
36.510	154.500	156.157	158.800
37.010	156.300	157.966	160.600
37.470	157.200	158.986	161.700

Layer velocity results

Layer	Top (m)	Bottom (m)	First peak Vs	First cross Vs	Max peak Vs	Selected residual (ms)
1	0.00	1.01	489.0	370.7	294.1	+1.068
2	1.01	1.50	385.9	313.0	226.7	+0.077
3	1.50	2.00	325.2	285.1	182.4	-1.212
4	2.00	2.50	289.4	273.8	161.2	-3.003
5	2.50	3.01	244.5	243.2	159.5	+2.465
6	3.01	3.51	177.4	177.2	169.2	+3.232
7	3.51	4.00	132.2	129.9	183.5	-2.208
8	4.00	4.51	131.8	129.1	195.4	-0.833
9	4.51	5.00	157.6	155.5	203.2	-0.057
10	5.00	5.50	194.1	193.2	210.6	+0.316
11	5.50	6.01	227.2	227.3	220.0	+0.168
12	6.01	6.51	249.8	249.4	231.1	+0.172
13	6.51	7.00	260.9	258.4	241.3	+0.038
14	7.00	7.51	264.0	259.4	248.6	+0.001
15	7.51	8.00	263.1	257.3	253.3	-0.036
16	8.00	8.51	260.0	255.6	256.3	-0.050
17	8.51	9.00	256.7	255.7	258.4	+0.102
18	9.00	9.51	254.9	257.8	260.2	-0.073
19	9.51	10.00	257.4	262.1	263.3	-0.067
20	10.00	10.51	261.2	266.7	266.5	-0.045
21	10.51	11.00	263.4	268.6	267.4	-0.108
22	11.00	11.51	261.8	265.7	264.1	+0.199
23	11.51	12.00	255.7	257.8	256.1	+0.113
24	12.00	12.51	248.2	249.0	247.5	-0.034
25	12.51	13.01	242.6	242.8	242.2	-0.156
26	13.01	13.53	239.5	239.6	239.6	+0.044
27	13.53	14.00	238.0	238.1	237.9	+0.199
28	14.00	14.51	238.5	238.6	237.4	-0.255
29	14.51	15.00	241.8	241.9	239.5	+0.016
30	15.00	15.50	245.3	244.8	242.0	+0.032
31	15.50	16.00	247.2	245.9	243.2	+0.102
32	16.00	16.51	246.9	245.0	243.0	-0.138
33	16.51	17.01	244.8	242.6	241.6	-0.014
34	17.01	17.51	239.6	237.8	237.3	+0.092
35	17.51	18.00	232.1	231.2	230.8	+0.137
36	18.00	18.51	224.9	224.8	224.5	-0.236
37	18.51	19.00	219.3	220.4	220.1	+0.116
38	19.00	19.51	214.4	216.6	216.3	-0.113
39	19.51	20.01	211.3	213.8	213.2	+0.228

Layer	Top (m)	Bottom (m)	First peak Vs	First cross Vs	Max peak Vs	Selected residual (ms)
40	20.01	20.51	210.4	211.6	210.9	-0.143
41	20.51	21.00	212.1	211.6	210.8	-0.128
42	21.00	21.51	213.8	212.3	211.2	+0.139
43	21.51	22.00	213.6	212.1	210.8	+0.031
44	22.00	22.50	213.3	212.2	211.1	-0.046
45	22.50	23.00	213.5	213.5	212.7	-0.024
46	23.00	23.51	215.2	215.9	215.2	+0.141
47	23.51	24.00	219.0	219.6	218.9	-0.092
48	24.00	24.50	225.4	225.2	224.1	-0.089
49	24.50	25.00	231.8	230.9	229.3	-0.036
50	25.00	25.51	234.8	234.2	232.3	-0.023
51	25.51	26.00	233.3	233.9	231.7	+0.184
52	26.00	26.50	229.6	231.0	228.7	+0.048
53	26.50	27.00	228.2	229.2	227.1	-0.057
54	27.00	27.51	230.9	230.8	228.5	+0.083
55	27.51	28.00	236.8	235.8	233.3	-0.196
56	28.00	28.51	244.7	243.3	240.8	-0.085
57	28.51	29.01	251.0	249.1	246.8	+0.125
58	29.01	29.51	252.8	250.2	248.1	-0.076
59	29.51	30.00	250.1	246.9	244.6	+0.100
60	30.00	30.52	242.9	239.8	237.3	+0.043
61	30.52	31.01	234.5	231.8	229.1	-0.001
62	31.01	31.50	227.5	225.2	222.5	-0.126
63	31.50	32.00	222.8	221.4	219.0	+0.172
64	32.00	32.51	220.9	220.6	218.5	-0.014
65	32.51	33.00	224.1	224.9	223.3	+0.010
66	33.00	33.50	233.6	235.0	233.5	-0.109
67	33.50	34.00	248.9	249.8	248.1	-0.032
68	34.00	34.51	266.8	266.3	263.7	-0.077
69	34.51	35.01	283.2	281.2	277.2	-0.030
70	35.01	35.51	294.7	291.6	286.3	+0.160
71	35.51	36.01	301.1	297.4	291.5	+0.187
72	36.01	36.51	306.4	302.2	296.3	-0.066
73	36.51	37.01	315.4	309.8	303.9	-0.266
74	37.01	37.47	328.1	320.5	314.2	+0.144